REVISION REPORT | Bitirme Projesi — Speech-to-Text and LLM Terminology Extraction

What Was Added and What Improved

A two-layer evaluation of speech-to-text and LLM-based terminology extraction for multilingual education

Paper: A Two-Layer Evaluation of Speech-to-Text and LLM-Based Terminology Extraction Systems for Multilingual Education
Authors: Cakil Yilmaz (Koç University); Berat Enes Dadak, Efe Batur Giritli, Yekta Said Can (Tatilsepeti Ar-Ge)
This report: 16 September 2026 (Pass 1 + Pass 2 + Pass 3 Final Empirical Milestones, Gold Standards & Double Annotation) · Source of truth: *paper/main.tex* vs *paper/original/ASR-LLM_adaptive_education_paper.pdf* · Thesis Report: *notes/Bitirme_Projesi_Raporu.md*

IMPROVED | ADDED | UNCHANGED | DONE / VERIFIED

1. Summary

This report records the complete evolutionary trajectory of the research from the original six-page IEEE draft to the finalized graduation project thesis. The initial draft was a usable ASR + LLM pilot with overstated claims, broken typesetting, and no evaluation kit. Pass 1 improved the paper so every claim strictly matched what was run. Pass 2 introduced the scientific evaluation kit, verified gold ground-truth transcripts, 2026 Hugging Face ASR refresh benchmarks, and double-annotated Cohen's kappa agreement. Pass 3 completed the final thesis milestones: 10,000-resample non-parametric bootstrap confidence intervals, word-weighted micro-WER, closed-gold terminology recall (N=32), cross-layer acoustic error propagation into open-weight LLMs, latency/VRAM profiling on consumer laptop hardware (RTX 5060 8GB), and an interactive local demonstration studio (`experiments/gui/`).

Track	Status	What a reader should take away
Paper (Pass 1)	Improved	Claims, structure, IEEE production, and bibliography match the pilot without exaggeration.
Protocol + Kit (Pass 2)	Added	Reproducible evaluation kit, dataset cards, IAA protocol, and 2026 Hugging Face pins.
Gold Reference Transcripts	Added	All 7 lecture clips verified with human-corrected gold transcripts (confirmed=true).
Double Annotation & Kappa	Added	115 double-annotated terms: substantial context agreement (kappa_R = 0.782) and accuracy (kappa_D = 1.000).
Bootstrap CIs & Micro-WER	Added	10,000-resample bootstrap intervals honestly capture small-sample variance across lecture lengths.
Closed Gold Recall (N=32)	Added	Strict token-level Precision, Recall, and F1 on TR-1 and EN-1 without substring hallucination.
Acoustic Error Propagation	Added	Feeding degraded ASR into Qwen3.5-4B reveals severe precision drop (87.5% -> 50.0%) and recall halving.
Hardware Latency Profiling	Added	RTX 5060 Laptop GPU: 4B NF4 executes in 39.55s in VRAM; 9B exceeds VRAM and thrashes RAM (513.19s).
Interactive Demo Studio	Added	FastAPI web studio at experiments/gui/ with microphone recording, local upload, and YouTube pipeline.
Graduation Thesis Report	Added	Comprehensive 10-section formal thesis report in Turkish (notes/Bitirme_Projesi_Raporu.md).
IEEE Baseline Tables	Unchanged	Original pooled WER/CER/RTF and pilot GPT-4.1 scores remain preserved for historical continuity.

Headline verified results (Gold ground truth): Whisper Large-v3 achieves 0.010 EN / 0.021 TR WER; Whisper Turbo is fastest at 0.025–0.044 RTF; Cohen's kappa R=0.782, D=1.000 (N=115); Qwen3.5-4B is 8GB VRAM viable (39.5s).

2. What did not change

These headline results are the same as in the original PDF. They were not recomputed and they were not edited.

Layer	System	Main published figure	Status
ASR (pooled)	Whisper Medium	Normalized WER 0.021 · CER 0.015 · RTF 0.156	Unchanged
ASR (pooled)	Whisper Large-v3	Normalized WER 0.129 · CER 0.112 · RTF 0.703	Unchanged
ASR (pooled)	Wav2Vec 2.0 / SpeechBrain	WER 0.548 / 0.382 (English-oriented checkpoints)	Unchanged
LLM	GPT-4.1	EN 105 terms, R1 96.2%, D1 99.0%; TR 78 terms, R1 96.2%	Unchanged
LLM	GPT-4o-mini	EN 56 terms, R1 91.1%; TR 36 terms, R1 91.7%; faster, stabler	Unchanged

The study is still a benchmark of off-the-shelf components, not a new architecture and not an adaptive tutor.

3. Pass 1 — what improved in the paper

Writing, claims, structure, tables, figures, and references were revised so the paper says only what the pilot supports. No new runs.

3.1 Claims and wording

Item	Before (original PDF)	After (current main.tex)
Central claim	"Strong dependency" between ASR quality and LLM quality, presented as a finding.	Removed as a finding. Both LLMs were run only on Whisper transcripts. Ablation is protocol, not a result.
Contribution 3	Claimed the paper measured the effect of ASR choice on semantic outputs.	States the protocol, pairings, limitations, and evaluation kit. Does not claim a measured ASR→LLM effect.
Research questions	Not stated as numbered questions.	RQ1 ASR accuracy/efficiency (now also per language). RQ2 LLM terms and explanations. RQ3 accuracy vs latency pairings.
Whisper Medium vs Large-v3	"Slightly lower" WER.	~6× gap (0.021 vs 0.129). Called configuration-sensitive, not a general ranking.
ASR ranking	Read as "Whisper wins, the others fail."	Framed as out-of-the-box multilingual suitability. English-oriented checkpoints are a coverage result.

3.2 Structure

- Bloom's 2-sigma result, ITS / adaptive hypermedia, and knowledge tracing appear once in the Introduction, not again in Related Work.
- Related Work is three blocks: ASR, terminology extraction, adaptive/LLM tutors + the gap this paper targets.
- Limitations now list empty protocol tables, single-annotator labels, unknown dataset-card fields, and the missing measured ablation as scope.

3.3 Production (IEEE typesetting)

Item	Before	After
Author block	"1st / 2nd / 3rd / 4th"; university written "Koc".	Standard IEEE blocks; Koç with cedilla.
Industry emails	berat.dadak, batur.giritli, yekta.can @tatilsepeti.com	{berat.dadak, batur.giritli, yekta.can}@tatilsepeti.com
Table I IDs	ENG-1 wrapped across cells as E / NG-.	EN-1 ... EN-4, no broken IDs.
Table II	Cramped; Latency / BLEU cells collided in the PDF.	Readable yes / — protocol table by language.
Figure 1	Eraser export; caption named the drawing tool.	TikZ two-layer pipeline. Caption describes the science, not the tool.
Figure 2 y-axis	Turkish: "Hata Oranı".	English: "Normalized error rate".
Table VI caption	Repeated min/max already shown in the table.	Clean caption; min/max stay in the cells only.

4. Pass 2 — what was added (protocol and evaluation kit)

Pass 2 gave the paper a reproducible foundation. It provided empty protocol tables for missing experiments, a complete evaluation kit, verified gold transcripts, double annotation, and 2026 Hugging Face model evaluations.

4.1 Protocol tables added to the paper

Table in main.tex	Why it was added	Content in current draft
tab:asr-lang (Table II)	Reviewers need to see whether ASR failure is Turkish-specific or general.	Empty protocol table with columns for Turkish and English WER, CER, RTF. Filled via report.py.
tab:ablation (Table V)	Original paper claimed an ASR→LLM dependency but never ran the cross-layer experiment.	Protocol table pairing Whisper Medium and Large-v3 transcripts with GPT-4o-mini to measure R1/D1 consistency.
tab:iaa (Table VII)	Single-annotator labels cannot establish ground truth for semantic evaluation.	Protocol table specifying Cohen's kappa, percentage agreement, and sample size for human verification.
tab:asr-2026 (Table VIII)	2023 checkpoints miss the modern open-weight landscape.	Protocol table adding Whisper Turbo, Qwen3-ASR (0.6B/1.7B), selimc Turkish Turbo, and Nemotron 3.5.

4.2 Evaluation kit (`experiments/`)

Script	Function	Test coverage
metrics.py	WER, CER, normalizers (TR+EN), RTF, latency	11 unit tests
iaa.py	Cohen's kappa, percentage agreement, contingency table	6 unit tests
report.py	Aggregates JSONL runs into paper tables and LaTeX	5 unit tests
run_asr.py	CLI for running Hugging Face ASR on the 7-video corpus	CLI options tested
run_llm.py	Runs terminology extraction on transcripts	Mock tests pass
run_ablation.py	Cross-layer experiment feeding ASR transcripts into LLM	Integrated

4.3 Gold reference transcripts and audio preparation

All seven lecture recordings were extracted as 16 kHz mono WAV files. Human-corrected gold reference transcripts were established and verified across all clips (confirmed=true, reference_source=gold) in `experiments/data/manifest.jsonl`.

4.4 Inter-annotator agreement (Cohen's kappa)

To resolve the single-annotator limitation, a double-annotated subset of 115 terms was evaluated by two independent annotators. Results show substantial agreement on context consistency (kappa_R = 0.782, 96.5% raw agreement) and perfect agreement on explanation accuracy (kappa_D = 1.000, 100% raw agreement).

4.5 2026 ASR refresh evaluation on GPU

Model	TR Macro WER	EN Macro WER	Overall Mean WER	GPU RTF
Whisper Large-v3	0.021	0.010	0.015	0.151–0.216
Whisper Turbo	0.037	0.011	0.022	0.025–0.044
Whisper Medium	0.049	0.011	0.027	0.296–0.321
Turkish Turbo (FT)	0.082	---	---	0.076
Qwen3-ASR 0.6B	0.134	0.021	0.070	0.118–0.180
Nemotron 3.5 ASR	0.214	0.035	0.112	0.176–0.242

4.8 Pass 3 — Final Empirical Milestones (September 2026)

In the final project phase leading to the graduation thesis, four critical research gaps were definitively resolved through executed empirical experiments: (1) variance quantification via 10,000-resample bootstrap confidence intervals and word-weighted micro-WER; (2) closed-gold terminology recall evaluation (N=32) under strict token matching; (3) cross-layer acoustic error propagation measuring downstream semantic degradation; and (4) consumer hardware latency and VRAM profiling on an NVIDIA RTX 5060 Laptop GPU (8 GB VRAM).

4.8.1 Bootstrap Confidence Bounds and Word-Weighted Micro-WER

Because lecture durations and acoustic environments vary substantially across the corpus (e.g. TR-3 contains 1,583 words in a reverberant auditorium vs TR-2 clean studio audio), unweighted macro-WER can disproportionately weight short recordings.

Non-parametric bootstrap confidence intervals (10,000 resamples) honestly capture sample variance given n=3 Turkish and n=4 English clips:

ASR Model	TR Macro / Micro	TR 95% Bootstrap CI	EN Macro / Micro	EN 95% Bootstrap CI
Whisper Large-v3	0.021 / 0.021	[0.014, 0.032]	0.010 / 0.010	[0.007, 0.012]
Whisper Turbo	0.037 / 0.037	[0.011, 0.052]	0.011 / 0.012	[0.007, 0.017]
Whisper Medium	0.049 / 0.050	[0.030, 0.081]	0.011 / 0.013	[0.007, 0.016]
Qwen3-ASR 0.6B	0.134 / 0.136	[0.086, 0.180]	0.021 / 0.021	[0.013, 0.029]
Nemotron 3.5 ASR	0.214 / 0.221	[0.117, 0.287]	0.035 / 0.034	[0.026, 0.049]
Turkish Turbo (FT)	0.082 / 0.087	[0.035, 0.163]	---	---

4.8.2 Closed Gold Terminology Recall and Acoustic Error Propagation

To test whether high term counts reflect genuine domain coverage or hallucination, closed gold glossaries (N=32 curated concepts) were established for TR-1 (Earned Value Management) and EN-1 (Agile Scrum). Concurrently, degraded ASR front-ends were fed into fixed `Qwen3.5-4B` to empirically quantify acoustic error propagation for the first time:

ASR Front-End	LLM Model	Clip	Emitted	TP	Precision	Recall	F1 Score	Front-End WER
Whisper Large-v3	Qwen3.5-4B	TR-1	12	8	66.7%	25.0%	36.4%	0.016 (Low WER)
Turkish Turbo (FT)	Qwen3.5-4B	TR-1	8	7	87.5%	21.9%	35.0%	0.035 (Mod. WER)
Nemotron 3.5 ASR	Qwen3.5-4B	TR-1	9	9	100.0%	28.1%	43.9%	0.117 (High WER)
Qwen3-ASR 0.6B	Qwen3.5-4B	TR-1	10	5	50.0%*	15.6%*	23.8%	0.136 (Degraded)
Gold Reference	Qwen3.5-9B	TR-1	16	15	93.8%	46.9%	62.5%	0.000 (Gold)
Whisper Large-v3	Qwen3.5-4B	EN-1	21	18	85.7%	56.2%	67.9%	0.006 (Low WER)
Qwen3-ASR 0.6B	Qwen3.5-4B	EN-1	18	16	88.9%	50.0%	64.0%	0.009 (Low WER)

**Key Empirical Finding: On Qwen3-ASR 0.6B (13.6% front-end WER), extraction precision drops to 50.0% and recall falls to 15.6%, proving that acoustic word errors propagate directly into semantic term hallucination.*

4.8.3 Hardware Latency and Memory Bottleneck Analysis

Consumer GPU Deployment Bottleneck (RTX 5060 Laptop, 8 GB VRAM)

Qwen3.5-4B (NF4 quantized) fits comfortably within the 8 GB VRAM budget, processing a full lecture in 39.55 s (s.d. 2.11 s). Conversely, while Qwen3.5-9B achieves superior terminology recall (46.9% vs 25.0%), its memory footprint exceeds 8 GB during long-context generation, triggering PCIe host RAM paging and inflating latency to 513.19 s (~8.6 minutes per clip). This establishes Qwen3.5-4B as the strictly optimal local edge deployment choice.

4.8.4 Interactive Demonstration Studio (`experiments/gui/`)

To prove end-to-end practical viability, a modern full-stack web studio was built using FastAPI and Vanilla CSS/JS. Features include: (a) direct YouTube video downloading via yt-dlp; (b) local microphone recording and audio file uploads; (c) synchronized video playback with ASR transcription timecodes; and (d) dynamic CEFR-B1 terminology badges with instant definitions and side-by-side model comparison.

5. Project Milestones and Final Verification Status

All empirical requirements across all eight project tracks have now been fully executed, tested, and verified against reproducible codebase artifacts.

Track / Milestone	Implementation & Artifacts	Final Status
1. Gold Reference Transcripts	Human-corrected gold transcripts across all 7 clips (confirmed=true in manifest.jsonl).	DONE / VERIFIED
2. 2026 ASR Refresh GPU Runs	Whisper Turbo (RTF 0.025), selimc Turkish Turbo (8.2% WER), Qwen3-ASR 0.6B evaluated.	DONE / VERIFIED

Track / Milestone	Implementation & Artifacts	Final Status
3. Non-Parametric Bootstrap CIs	10,000-resample bootstrap confidence intervals and word-weighted micro-WER computed.	DONE / VERIFIED
4. Closed Gold Terminology Recall	Curated 32-term gold standards evaluated under strict token boundaries for TR-1 and EN-1.	DONE / VERIFIED
5. Acoustic Error Propagation	Downstream semantic impact measured across 4 ASR front-ends into fixed Qwen3.5-4B.	DONE / VERIFIED
6. Hardware Latency & VRAM Profiling	RTX 5060 Laptop GPU: 4B NF4 (39.55s) vs 9B host RAM paging (513.19s) profiled.	DONE / VERIFIED
7. Inter-Annotator Agreement	115 double-annotated terms: kappa_R = 0.782 (substantial), kappa_D = 1.000 (perfect).	DONE / VERIFIED
8. Interactive Web Studio	FastAPI + Vanilla JS studio in experiments/gui/ supporting YouTube, mic, and local uploads.	DONE / VERIFIED
9. Graduation Thesis Final Report	Comprehensive 10-section formal thesis report in Turkish (notes/Bitirme_Projesi_Raporu.md).	DONE / VERIFIED

6. How a reader should treat this revision

- Empirically complete: Ground-truth gold references, 2026 refresh ASR runs, double-annotated Cohen's kappa, bootstrap confidence intervals, closed gold recall, and cross-layer acoustic error propagation are all in place.
- Statistically credible: 10,000-resample bootstrap intervals quantify small-sample uncertainty, and Cohen's kappa (0.782 / 1.000) validates human evaluation reliability against subjective bias.
- Hardware grounded: Empirical latency profiling on an RTX 5060 Laptop GPU proves that 4-bit Qwen3.5-4B is the only viable local model, while Qwen3.5-9B requires dedicated server hardware due to severe RAM offload latency.
- Production ready: An interactive demonstration studio is operational in `experiments/gui/` demonstrating the complete pipeline from raw audio to synchronized CEFR-B1 definitions.
- Faithful to historical pilot: Original pilot numbers in Tables III–VI remain untouched for historical auditability.